# Gemini API + YouTube Data API

What they are, what they can do, app ideas, and documentation | Prepared for Tushara Hall | November 2, 2026

## At a glance

| | Gemini API | YouTube Data API v3 |
|---|---|---|
| What it is | Google's AI model API. It reads and writes text, and understands images, video, audio and PDFs. | Google's API for YouTube's catalog. It searches and retrieves videos, channels, playlists and comments. |
| Think of it as | The **brain**: understands, summarizes, reasons, writes | The **library**: finds and plays the right video |
| Where you get a key | Google AI Studio (aistudio.google.com) | Google Cloud Console (APIs & Services) |
| Free? | Yes: free tier with rate limits, no credit card | Yes: 10,000 quota units per day |
| Main catch | Free-tier inputs are used to improve Google products; limits can change | A search costs 100 units, so roughly 100 searches/day |

**Why they pair well:** Gemini can take a **public YouTube URL directly** and summarize it, answer questions about it, transcribe it, or point to timestamps. YouTube finds the videos; Gemini understands them.

## Gemini API: capabilities

Free-tier models include Gemini 3.8 Flash (the model used in Google's current examples), Gemini 3.6 Flash and Gemini 3.5 Flash-Lite. Image, video and music generation models are **paid only**.

| Capability | What you can do | Free tier? | Docs |
|---|---|---|---|
| Text generation & chat | Answer questions, write, summarize, run multi-turn chat | Yes | Text generation |
| Image understanding | Identify objects, read text in photos, describe or compare images | Yes | Image understanding |
| Video understanding | Summarize, Q&A, timestamps; accepts **YouTube URLs** (public only; free tier max 8 hours of YouTube video/day) | Yes | Video understanding |
| Audio understanding | Transcribe, summarize or translate audio; speaker labels | Yes | Audio |
| Document (PDF) processing | Read PDFs including tables and charts; extract or summarize | Yes | Document processing |
| Structured output | Force replies into JSON your app can use (e.g. an ingredient list) | Yes | Structured output |
| Function calling | Let Gemini call your own code or other APIs (e.g. YouTube search) | Yes | Function calling |
| Grounding with Google Search | Answers backed by live web results with citations | Limited | Google Search grounding |
| URL context | Read web pages you pass in and use them in the answer | Yes | URL context |

| Capability | What you can do | Free tier? | Docs |
|---|---|---|---|
| Code execution | Gemini writes and runs Python for math and data tasks | Yes | Code execution |
| Embeddings | Turn text, images, video or PDFs into vectors for search and recommendations | Yes | Embeddings |
| Long context | Work with very large inputs (up to about 1M tokens on many models) | Yes | Long context |
| Live API | Real-time voice conversations and live translation | Yes | Live API |
| Speech generation (TTS) | Turn text into natural-sounding speech | Yes | Speech generation |
| Image generation (Nano Banana) | Create or edit images from prompts | **Paid** | Image generation |
| Video & music generation | Gemini Omni (video) and Lyria (music) | **Paid** | Pricing |

## YouTube Data API v3: capabilities

Every request costs quota units from your 10,000 per day. Reading details is cheap; searching is expensive.

| Resource / method | What you get | Quota cost | Docs |
|---|---|---|---|
| search.list | Find videos, channels or playlists by keyword, with filters (length, date, region, channel) | 100 | search.list |
| videos.list | Title, description, duration, views, likes, comment count, tags, thumbnails | 1 | videos.list |
| channels.list | Channel name, subscribers, total views, video count | 1 | channels.list |
| playlists / playlistItems.list | All videos in a playlist or a channel's uploads | 1 | playlistItems.list |
| commentThreads.list | Public comments and replies on a video | 1 | commentThreads.list |
| Write actions | Create playlists, add videos, rate, comment (user must sign in with Google/OAuth) | 50+ | API reference |
| IFrame Player (separate) | Embed and control a video player in your page | Free, no key | IFrame Player API |

**What it can't do:** download or re-host videos, give you full transcripts of other people's videos, or tell you what happens inside a video. (Gemini covers that last part.)

## App ideas

All of these can be built for $0 on Netlify using free-tier Gemini models and YouTube's free quota.

| App | What it does | APIs used |
|---|---|---|
| **Fridge Photo Chef** | Snap a fridge photo; Gemini lists the ingredients; YouTube shows recipe videos using them | Gemini image + structured output, YouTube search |
| **Video-to-Study-Notes** | Paste a lecture or tutorial link; get a summary, key terms and a quiz | Gemini video (YouTube URL) + structured output |

| App | What it does | APIs used |
|---|---|---|
| **Timestamp Finder** | Ask "where does she explain X?" and jump straight to that moment | Gemini video + YouTube IFrame Player |
| **Comment Sentiment Dashboard** | Pull a video's comments and chart positive vs. negative themes. Strong Business Analytics project | YouTube commentThreads + Gemini |
| **Product Review Summarizer** | Enter a product; find review videos; Gemini lists pros, cons and a verdict | YouTube search + Gemini video |
| **Competitor Channel Analyzer** | Compare channels' subscribers, views and posting cadence, plus AI summaries of top videos | YouTube channels/videos + Gemini |
| **Recipe Card Extractor** | Turn any cooking video into a written recipe card and shopping list | Gemini video + structured output |
| **Travel Vlog Planner** | Search vlogs for a city; Gemini pulls out the places mentioned into an itinerary | YouTube search + Gemini video |
| **Language Practice from Videos** | Pick a video in Spanish or French; get a transcript, translation and vocab list | Gemini video/audio + YouTube search |
| **Podcast-to-Blog** | Turn a YouTube podcast episode into a blog post with quotes and timestamps | Gemini video + YouTube videos.list |

## How to build on Netlify (free)

- Get a Gemini key in Google AI Studio and a YouTube key in Google Cloud Console.
- Store both keys as Netlify environment variables. Never put them in front-end code.
- Make API calls from a Netlify Function so the keys stay hidden.
- Cache YouTube results for repeat searches to stay under 10,000 units/day.
- Check the Gemini rate limits page before a demo; free-tier limits change.
- Don't send private photos or data on the free tier, since Google may use that content to improve its products.

## Documentation links

### *Gemini API*

- Docs home
- Quickstart
- Get an API key
- Models
- Pricing & free tier
- Rate limits
- Video understanding (YouTube URLs)
- Image understanding
- Structured output
- Function calling

### *YouTube Data API v3*

- Getting started

- API reference
- search.list
- videos.list
- commentThreads.list
- Quota costs
- IFrame Player API

### *Netlify*

- Netlify Functions
- Environment variables

Sources checked November 2, 2026: Gemini API pricing and video-understanding pages; YouTube Data API overview and quota-cost pages.